Extra Bite #3: Semiconductor Device Recognition

Diodes, transistors, FETs: what they are made of and how they behave. One pass over E6A's 12 questions.

Group E6A · 12 pool questions · Standalone · 5-10 minute presentation · Practice exams at eham.net/exams

Why You Care

Every radio is built on a handful of semiconductor devices. Extra does not ask you to design them; it asks you to recognize them, name their materials, and know one or two numbers about each. This Bite is high-yield because it is mostly pattern recognition: once you know the vocabulary, the questions answer themselves.

What "Semiconductor" Actually Means

A pure silicon (or germanium) crystal does not conduct well. To make it useful, manufacturers add a tiny amount of a different element. This is called **doping**. There are two flavors:

Type	What it has extra of	Impurity name
N-type	Free **electrons** (negative charge carriers)	**Donor** impurity (donates electrons)
P-type	"Holes": missing electrons that act like positive carriers	**Acceptor** impurity (accepts electrons, creates holes)

So "which semiconductor material contains excess free electrons?" → **N-type**. "The impurity that adds holes" → **acceptor**. (E6A02, E6A04)

One Special Material: Gallium Arsenide

Silicon is the workhorse, but it gets slow at microwave frequencies. **Gallium arsenide (GaAs)** works much faster, so it is the material of choice for **microwave** devices (GaAsFETs in satellite preamps, etc.). (E6A01)

The PN Junction Diode

Push P-type and N-type material together and you get a **PN junction**: a diode. At the boundary, electrons from the N side and holes from the P side combine, leaving a thin neutral layer called the **depletion region**.

- **Forward bias** (positive on P, negative on N): pushes carriers across the junction. The diode conducts.
- **Reverse bias** (positive on N, negative on P): pulls carriers **away** from the junction. The depletion region widens. No current flows. (E6A03)

Bipolar Junction Transistors (BJTs)

A BJT is two PN junctions back to back: **NPN** or **PNP**. Three pins: base, collector, emitter. A small base current controls a much larger collector current.

Beta: change in collector current ÷ change in base current. If +1 mA on the base causes +100 mA on the collector, beta = 100. (E6A06)
Vbe (silicon, biased on): base-to-emitter **voltage** of about **0.6 to 0.7 V**. That is the diode drop of the base-emitter junction. (E6A07)
Alpha cutoff frequency: the frequency at which the grounded-base current gain has dropped to **0.7** of its 1 kHz value. Standard speed spec. (E6A08)

Field-Effect Transistors (FETs)

A FET swaps the BJT's base current for an insulated **gate voltage** that controls source-to-drain current. Because the gate is insulated, almost no DC current flows into it – so a FET's DC input impedance is **much higher** than a BJT's. (E6A05)

Depletion-mode FETs have a channel by default – current flows source-to-drain at **zero gate voltage**, and the gate is used to pinch the channel off (E6A09). Enhancement-mode FETs are the opposite: no channel until the gate is driven.

One Protection Trick: Zeners on a MOSFET Gate

What a Zener is, first: A Zener diode is a regular PN-junction diode that has been engineered to "break down" at a precise **reverse** voltage and conduct in that direction, holding the voltage steady at that breakdown value. Forward-biased, it acts like an ordinary diode (current flows). Reverse-biased above its rated voltage, it conducts and clamps the voltage to that rated number. This makes Zeners useful both as **voltage references** (regulators) and as **over-voltage clamps** (protection).

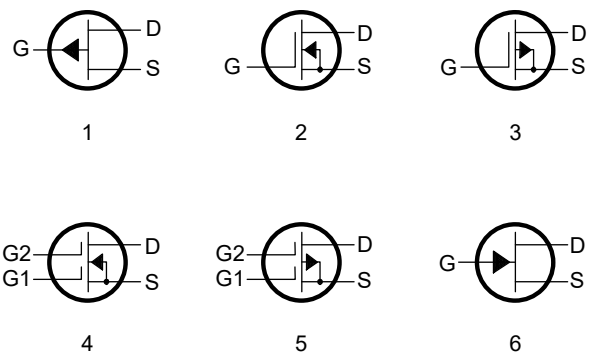
MOSFET gates are insulated from the rest of the device by an extremely thin oxide layer. A static-electricity zap punches through that oxide and destroys the part. So MOSFETs are routinely built with small **Zener diodes connected from gate to source (or drain)** that clamp any over-voltage and shunt static energy to ground.

Zeners on a MOSFET gate = static (ESD) protection. (E6A12)

Schematic Symbols (Official Figure E6-1)

This is the figure as it appears on the exam. Questions E6A10 and E6A11 refer to **this** diagram and its numbered symbols.

Figure E6-1



The letters next to each symbol are the device's terminals. On a FET they are:

- **G** (Gate) – the control input. Voltage here sets up the electric field that controls source-to-drain current. On dual-gate MOSFETs there are two of these, labeled **G1** and **G2**.
- **D** (Drain) – one end of the channel. Current normally flows in here (for an N-channel FET).
- **S** (Source) – the other end of the channel. Current normally flows out here (N-channel).

Cues in the symbol:

Cue	What it tells you
Gate arrow points **in** toward channel	N-channel
Gate arrow points **out** away from channel	P-channel
Two separate gate lines (G1, G2)	Dual-gate MOSFET
Solid channel line	Depletion-mode
Broken (dashed) channel line	Enhancement-mode

The figure shows six numbered FET symbols. The **top row** (1, 2, 3) are **single-gate FETs** (one G, one D, one S). The **bottom row** contains symbols 4 and 5 (dual-gate MOSFETs with G1 and G2) plus symbol 6 (a single-gate device).

The 2024-2028 Extra pool tests only two specific identifications from this figure. Memorize these:

Symbol	What it is	Exam Q
1	**P-channel JFET** (single gate, arrow on gate points **away** from the channel = P-channel)	E6A11
4	**N-channel dual-gate MOSFET** (two gate lines G1 and G2, gate arrow points **in** toward channel = N-channel)	E6A10

Symbols 2, 3, 5, and 6 are other FET variations shown in the figure for completeness but **not tested in the current pool**. Apply the cues above (count the gate lines, watch the arrow direction) to identify any FET symbol without memorizing all six.

- **N-channel dual-gate MOSFET:** two gate inputs with channel arrow in. In E6-1 this is symbol **4**. (E6A10)
- **P-channel junction FET:** single gate with arrow out. In E6-1 this is symbol **1**. (E6A11)

Memorize the **cues**, not the symbol numbers; if a future pool revision shuffles the figure, the cues still hold.

Mental Cheat Sheet

- N-type = extra electrons; donor impurity.
- P-type = extra holes; acceptor impurity.
- Gallium arsenide → microwave.
- Reverse-biased diode: depletion region widens, no current flows.
- BJT beta = change in Ic ÷ change in Ib.
- Silicon NPN biased on: Vbe ≈ 0.6 to 0.7 V.
- Alpha cutoff frequency = grounded-base gain drops to 0.7 of 1 kHz value.
- FET gate has very high DC input impedance vs BJT.
- Depletion-mode FET: current flows at zero gate voltage.
- Dual-gate MOSFET symbol: two gate inputs (G1, G2).
- JFET arrow points **in** for N-channel, **out** for P-channel.
- Zeners gate-to-source on a MOSFET: static protection.

Common Traps

- "Excess free electrons" sounds negative, but the answer is **N-type**, not P-type. Free electrons are negative carriers; that is what "N" stands for.
- The impurity-name pair is reversed from intuition: **donor** donates electrons (N-type); **acceptor** creates holes (P-type).
- The reverse-bias question wants the **depletion region widening**. Answers about "only P" or "only N" conducting are wrong.
- Beta is a **ratio of changes**, not the value of Ic itself.
- The Vbe answer is in **volts** (0.6 to 0.7 V), not ohms. The test offers a "0.6 to 0.7 ohms" trap.
- Zeners on a MOSFET gate protect from **static**, not from overheating or bias drift.

Takeaway: Doping creates N-type (electrons) and P-type (holes); diodes are one PN junction, BJTs are two; FETs swap base current for an insulated gate voltage; gallium arsenide is the microwave material; and a Zener on a MOSFET gate is there to stop static damage.

Bite 3 Question Bank (E6A)

12 questions. Read, decide, check the answer key at the end.

E6A01. In what application is gallium arsenide used as a semiconductor material?
A. In high-current rectifier circuits.
B. In high-power audio circuits.
C. In microwave circuits.
D. In very low-frequency RF circuits.

E6A02. Which of the following semiconductor materials contains excess free electrons?
A. N-type.
B. P-type.
C. Bipolar.
D. Insulated gate.

E6A03. Why does a PN-junction diode not conduct current when reverse biased?
A. Only P-type semiconductor material can conduct current.
B. Only N-type semiconductor material can conduct current.
C. Holes in P-type material and electrons in the N-type material are separated by the applied voltage, widening the depletion region.
D. Excess holes in P-type material combine with the electrons in N-type material, converting the entire diode into an insulator.

E6A04. What is the name given to an impurity atom that adds holes to a semiconductor crystal structure?
A. Insulator impurity.
B. N-type impurity.
C. Acceptor impurity.
D. Donor impurity.

E6A05. How does DC input impedance at the gate of a field-effect transistor (FET) compare with that of a bipolar transistor?
A. They are both low impedance.
B. An FET has lower input impedance.
C. An FET has higher input impedance.
D. They are both high impedance.

E6A06. What is the beta of a bipolar junction transistor?
A. The frequency at which the current gain is reduced to 0.707.
B. The change in collector current with respect to the change in base current.
C. The breakdown voltage of the base-to-collector junction.
D. The switching speed.

E6A07. Which of the following indicates that a silicon NPN junction transistor is biased on?
A. Base-to-emitter resistance of approximately 6 ohms to 7 ohms.
B. Base-to-emitter resistance of approximately 0.6 ohms to 0.7 ohms.
C. Base-to-emitter voltage of approximately 6 volts to 7 volts.
D. Base-to-emitter voltage of approximately 0.6 volts to 0.7 volts.

E6A08. What is the term for the frequency at which the grounded-base current gain of a bipolar junction transistor has decreased to 0.7 of the gain obtainable at 1 kHz?
A. Corner frequency.
B. Alpha rejection frequency.
C. Beta cutoff frequency.
D. Alpha cutoff frequency.

E6A09. What is a depletion-mode field-effect transistor (FET)?
A. An FET that exhibits a current flow between source and drain when no gate voltage is applied.
B. An FET that has no current flow between source and drain when no gate voltage is applied.
C. An FET that exhibits very high electron mobility due to a lack of holes in the N-type material.
D. An FET for which holes are the majority carriers.

E6A10. In Figure E6-1, which is the schematic symbol for an N-channel dual-gate MOSFET?
A. 2.
B. 4.
C. 5.
D. 6.

E6A11. In Figure E6-1, which is the schematic symbol for a P-channel junction FET?
A. 1.
B. 2.
C. 3.
D. 6.

E6A12. What is the purpose of connecting Zener diodes between a MOSFET gate and its source or drain?
A. To provide a voltage reference for the correct amount of reverse-bias gate voltage.
B. To protect the substrate from excessive voltages.
C. To keep the gate voltage within specifications and prevent the device from overheating.
D. To protect the gate from static damage.

Answer Key

E6A01: C	E6A02: A	E6A03: C
E6A04: C	E6A05: C	E6A06: B
E6A07: D	E6A08: D	E6A09: A
E6A10: B	E6A11: A	E6A12: D